

SYSTEMIC RISK IN FINANCIAL NETWORKS REVISITED

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FACTS

Banks' gross debts bigger than net

E.g. HSBC's net position $|\pounds 24\text{B} - \pounds 21.5\text{B}| \approx 10\%$ gross

Thought to harbor systemic risk $\implies$ Policy makers advocate netting out

Supported by networks models (e.g. Acemoglu–Ozdaglar–Tahbaz-Salehi 15)

Based on one-period debt capturing overnight debts (e.g. repos)

Much interbank debt longer maturity

Germany: Average mat. more than year; frac. overnight less than 10%

QUESTIONS

Do long-term debt networks harbor same systemic risks as short-?

Do the same network structures lead risks to propagate?

Do gross debts serve function that could be undermined by netting out?

THIS PAPER

Model of N banks connected in network of long-term debts

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Assumption: $y > \ell > \theta y$

LIMITED PLEDGEABILITY

B_i's Balance Sheet

Assets	Liabilities
long-term investments	short-term liq. shock
	equity

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y	short-term
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debt from B _j	debt to B _j
	equity

RESULTS

High indebtedness and connectedness sources of value and stability

Zero net long-term positions have positive NPV

Embed option to dilute with new debt $\implies$ liquidity insurance

Contingent transfers via plain debt

“Exponential networks” implement optimal transfers for any shocks

RESULTS MATTER FOR POLICY

Policies that help with short-term debt backfire with long-term debt

Decreasing indebtedness/connectedness can decrease efficiency

MODEL

MODEL OVERVIEW

Two dates: Date 1 and Date 2; no discounting; universal risk neutrality

N banks: Assets y at Date 2 and risk of liquidity shock $\ell < y$ at Date 1

Interbank network: Network of long-term debts $\mathbf{F} = [F_{i \rightarrow j}]_{ij}$ (due at Date 2)

Friction: Limited pledgeability: Only $\theta y < \ell$ pledgeable

BANKS $B_1, \dots, B_N$ AND BALANCE SHEETS

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B_i has pledgeable assets $\theta y + \text{PV}[F_{i \Leftarrow}]$

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B_i has pledgeable assets $\theta y + \text{PV}[F_{i \Leftarrow}]$

Assumption: New debt senior (e.g. repo) $\implies F_{i \Rightarrow}$ diluted

REPAYMENTS

Denote B_i 's equilibrium repayment to B_j by $R_{i \rightarrow j}$

Total repayments: $R_{i \Rightarrow} := \sum_j R_{i \rightarrow j}$ and $R_{i \Leftarrow} := \sum_j R_{j \rightarrow i}$

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NB: Liquidation inefficient (destroys $(1 - \theta)y$), default alone is not (transfer)

EQUILIBRIUM

A payment equilibrium is a repayment profile $[R_{i \rightarrow j}]_{ij}$ for each $(\sigma_i)_i$ s.t.

Repayments are sequentially rational

Repayments are paid pro rata: $\frac{R_{i \rightarrow j}}{R_{i \rightleftharpoons}} = \frac{F_{i \rightarrow j}}{F_{i \rightleftharpoons}}$

TIMELINE/SUMMARY

Date 1: Shocks realized; banks raise new liq.; banks liquidated/continue

Date 2: Assets y realized; banks repay or default

DEFINITION: EFFICIENCY

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A network more efficient than another if fewer banks liquidated $\forall (\sigma_i)_i$

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Benchmark isomorphic to AOT mutatis mutandis

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But claims more than fully encumbered by liabilities created on RHS

BM2: DEFAULT RADIUS

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Neighbors’ neighbors provide them liquidity...

Overall: Not-shocked near shocked pay out so much that can’t meet shocks

BM3: CONNECTEDNESS

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Formalized using “bottleneck parameter”/“delta connectedness”

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Intuition: Liquidations propagate through network per default radius (BM2)

RESULTS

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Claims not encumbered by liabilities created on RHS (can be diluted)

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B_i 's Balance Sheet

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long-term investments	short-term liq. shock
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B_i 's Balance Sheet

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B_i 's Balance Sheet

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SHOCKED BANK B_i NEEDS LIQUIDITY

B_i 's Balance Sheet

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debt from B_j	debt to B_j
	equity

SHOCKED BANK B_i NEEDS LIQUIDITY

B_i 's Balance Sheet

Assets	Liabilities
y	ℓ
αF	αF
	equity

B_i RAISES CASH VIA NEW DEBT AGAINST y & αF

B_i 's Balance Sheet

Assets	Liabilities
y	ℓ
αF	αF
	equity

$\rightarrow$

B_i 's Balance Sheet

Assets	Liabilities
y	ℓ
αF	αF
cash	new debt
	equity

DILUTES B_j

B_i 's Balance Sheet

Assets	Liabilities
y	ℓ
αF	αF
	equity

$\rightarrow$

B_i 's Balance Sheet

Assets	Liabilities
y	ℓ
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	equity

B_j NOT WORSE OFF EX ANTE

Gross debts mean B_j diluted when B_i is shocked

But B_j can also dilute B_i when it is shocked

Gross debt implement transfer from not-shocked to shocked bank

Coinsurance via option to dilute

PRACTICAL IMPLEMENTATION

Banks hold gross long-term dilutable debts

E.g. interbank loans/bonds

Rationalizes why long-maturity

Banks dilute with short-term senior debt

Rationalizes e.g. super-seniority for repos

Explains large interbank positions (quarter of balance sheets)

DILUTION COMPLEMENTS DEFAULT

Banks use the option to default to implement contingencies

Implements transfer from not-shocked to shocked at Date 2

Allen–Gale 98, Dubey–Geanakoplos–Shubik 88, and Zame 93

But default not enough here

Need dilution to prevent liquidation at Date 1

Like defaultable debt, dilutable debt can be good

Implements transfers before maturity

R2: SALVATION RADIUS

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Overall: Banks near not-shocked banks dilute so much that meet shocks

R3: CONNECTEDNESS INCREASES EFFICIENCY

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Increasing connectedness increases efficiency

Formalized using “bottleneck parameter”/“delta connectedness”

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Intuition: Liquidity propagates through network per salvation radius (R2)

TO SUM UP: LT DEBT NETWORKS UNLIKE ST

Indebtedness and connectedness sources of efficiency

Reason: Option to dilute gross debts provides insurance

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Answer: No!

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Indebtedness and connectedness sources of efficiency

Reason: Option to dilute gross debts provides insurance

Question: Do high indebtedness and connectedness suffice for efficiency?

Answer: No! Complete network (fully connected) inefficient no matter debt

R4: COMPLETE NETWORK INEFFICIENT

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Let S be number of shocked banks and suppose $S\ell > N\theta y$

If $\mathbf{F}$ is complete ($F_{i \rightarrow j} \equiv F$) then all shocked banks are liquidated

R4: COMPLETE INEFFICIENT: PROOF (SKETCH)

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R4: COMPLETE INEFFICIENT: INTUITION

Complete network delivers all shocked banks same net payment

If not enough to save all, each gets same insufficient amount of liquidity

None saved

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Question: How much better can we do?

DEFINITION: CONSTRAINED EFFICIENCY

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A network is constrained efficient if L is minimized for each $(\sigma_i)_i$ s.t.

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I.e. liq. provided to shocked not-liquidated $\leq$ available from not-shocked

PRINCIPLES OF EFFICIENCY

Planner should allocate liquidity to save largest number of shocked banks:

- (i) Not-shocked banks pay out all liquidity (θy)
- (ii) Allocate none to liquidated banks (so all used to save shocked)

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NB: Ordering by indices arbitrary, can consider permutation

R5: EXPONENTIAL NETWORKS ARE CONSTRAINED EFFICIENT

R5: EXP. NETWORKS CONSTRAINED EFFICIENT

Let $\mathbf{F}$ be an exponential network with base s small enough

For α large enough, $\alpha\mathbf{F}$ is generically constrained efficient

R5: EXP. NETWORKS EFF.: PROOF (SKETCH)

Echoes principles of efficiency

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CONCLUSION

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Off-setting long-term debts provide insurance

Indebtedness and connectedness sources of efficiency

Contrary to conclusions based on short-term debt

Indebtedness and connectedness implement efficiency if network exponential

Minimize number of liquidations no matter realization of shocks

“Robust but never fragile”

SYSTEMIC RISK IN FINANCIAL NETWORKS REVISITED

APPENDIX

NON-CONTINGENT LIQUIDITY

If transfer ℓ to all banks at Date 0 at rate $R = \frac{L - \pi\theta y}{(1 - \pi)L}$

All banks meet their shocks

Shocked banks repay θy , not-shocked banks repay RL

Outside lender breaks even

Works, but requires outside liquidity $NL \gg ML$ at Date 0

OUTSIDE CREDIT LINES

Extend credit line to all banks to borrow ℓ at Date 1 at rate ϵ

For non-contingent repayment $F = \frac{L - \pi\theta y}{1 - \pi}$

Shocked banks draw down, not shocked banks don't

Outside lender breaks even

Works, but requires commitment from outside lender